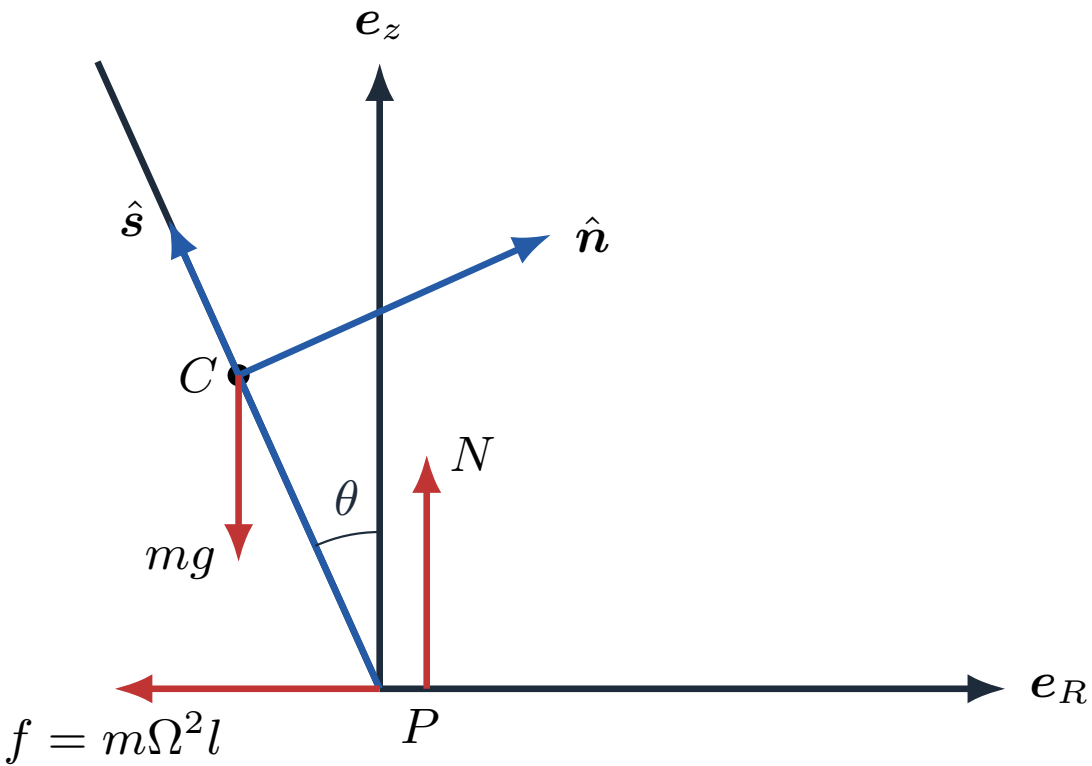

$$\omega_n = -\Omega l/r$$

$$\omega_x = -\Omega(l/r + \sin\theta)\cos\theta$$

$$\omega_y = \Omega(\cos^2\theta - l\sin\theta/r)$$

摩擦沿 $-\boldsymbol{e}_R$；接触力矩不为零